Yijia Fan

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I am seeking PhD opportunities for Fall 2027. If you are interested, please contact me directly! My WeChat ID can be found through my phone number.

Research Interests

My research is anchored in the continuous evolution of Visual Language Models (VLMs), synergizing them with the advanced agentic capabilities of Large Language Models (LLMs) to construct robust multi-modal intelligence. I focus on aligning visual perception with complex reasoning via reinforcement learning. Building on my experience with LLaVA architectures and post-training techniques like RFT, I concurrently explore pathways for unified generation. This involves leveraging semantic representations and VAE modules to bridge the gap between image understanding and video synthesis. Ultimately, my long-term objective is to forge a unified world model that seamlessly integrates perception, agentic reasoning, and generative creation. I aspire to bridge the gap between theoretical innovation and practical scalability, delivering high-impact contributions that resonate across both academic frontiers and industrial applications.

Education

Computer Science, SUN YAT-SEN University **Guangzhou, China**
B.Eng. in Computer Science and Technology *Sept. 2023 - June 2027 (Expected)*
GPA: 3.5/4.0
Relevant Courses: Mathematical Analysis, Linear Algebra, Artificial Intelligence, Data Structures and Algorithms, Operating System, Computer Architecture

Experience

SUN YAT-SEN University – Human Cyber Physical Intelligence Integration Lab **Guangzhou, China**
July. 2024 – Present
Undergraduate Research Intern Supervisor: Prof. Keze Wang

- Investigating diffusion-based methods to extend the context window and processing capabilities of LLMs for ultra-long texts.
- Developed a game-theoretic multi-agent system to optimize MAS.
- Implemented a curiosity-driven exploration mechanism to address few-shot classification challenges.

X-Era & SUN YAT-SEN University Pop Mart Enterprise Project **Guangzhou, China**
Undergraduate Research Intern Supervisor: Prof. Keze Wang May. 2025 – July. 2025

- Led the development of a Pop Mart product defect detection pipeline by combining VisionLanguage Models (VLMs) with Reinforcement Learning from Human Feedback (RLHF).
- Designed an active learning loop where VLM-generated annotations were iteratively refined through RLHF with expert reviewers.

Microsoft Research Asia Shanghai ML Group **Shanghai, China**
Research Intern Mentor: Senior Research SDE. Yifan Yang July. 2025 – Present

- Participate in research on improving video generation using hierarchical VAE.
- Spearheading the exploration for the next generation of discretized VLM projects.
- Contributed to the training of the Unified video generation and understanding model.
- Experienced in training on multi-node clusters and resolving DeepSpeed synchronization issues.

Conference Publications

[1] **[AAAI'26]** Yijia Fan, Jusheng Zhang, Kaitong Cai, Jing Yang, Chengpei Tang, Jian Wang, and Keze Wang. "Cost-Effective Communication: An Auction-based Method for Language Agent Interaction." [*Paper*]

[2] **[AAAI'26]** Yijia Fan, Jusheng Zhang, Kaitong Cai, Jing Yang, Jian Wang, and Keze Wang. "3DAlign-DAER: Dynamic Attention Policy and Efficient Retrieval Strategy for Fine-grained 3D-Text Alignment at Scale." [*Paper*]

[3] **[NeurIPS'25]** Jusheng Zhang, Yijia Fan, Wenjun Lin, Ruiqi Chen, Haoyi Jiang, Wenhao Chai, Jian Wang, and Keze Wang. "GAM-Agent: Game-Theoretic and Uncertainty-Aware Collaboration for Complex Visual Reasoning." [*Paper*]

[4] **[NeurIPS'25]** Jusheng Zhang, Yijia Fan, Zimo Wen, Keze Wang, and Jian Wang. "Tri-MARF: A Tri-Modal Multi-Agent Responsive Framework for Comprehensive 3D Object Annotation." [*Paper*]

[5] **[ICML'25]** Jusheng Zhang, Zimeng Huang, Yijia Fan, Ningyuan Liu, Mingyan Li, Zhuojie Yang, Jiawei Yao, Jian Wang, and Keze Wang. "KABB: Knowledge-Aware Bayesian Bandits for Dynamic Expert Coordination in Multi-Agent Systems." [*Paper*] [GitHub]

[6] **[EMNLP'25]** Yijia Fan, Jusheng Zhang, Kaitong Cai, Jing Yang, and Keze Wang. "CCG: Rare-Label Prediction via Neural SEMDriven Causal Game." [*Findings*]

[7] **[EMNLP'25]** Jusheng Zhang, Yijia Fan, Kaitong Cai, Zimeng Huang, Xiaofei Sun, Jian Wang, Chenpei Tang, and Keze Wang. "DrDiff: Dynamic Routing Diffusion with Hierarchical Attention for Breaking the Efficiency-Quality Trade-off." [*Paper*]

[8] **[EMNLP'25]** Jusheng Zhang, Yijia Fan, Kaitong Cai, Jingzhou Tang, Xiaofei Sun, and Keze Wang. "OSC: Cognitive Orchestration through Dynamic Knowledge Alignment in Multi-Agent LLM Collaboration." [*Findings*]

Academic Service

Conference Reviewing. **CVPR** 2026, **ICML** 2026, **ICLR** 2026, **AAAI** 2026, **ICME** 2025 (*Outstanding Reviewer*).